Final Project Proposal:
Breaking and Fixing Deepfake Detectors Under Adversarial Attacks

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Problem

Most deepfake detectors work well on clean test sets, but in practice, images go through JPEG compression, resizing, blurring, and other distortions before anyone even looks at them. Social media platforms compress and resize everything. On top of that, adversarial attacks like FGSM and PGD can make a detector misclassify with perturbations that are basically invisible to the human eye. I want to see how fragile a standard deepfake detector actually is under these conditions, and whether adversarial training can make it hold up better.

Approach

The plan is to fine tune a pretrained CNN on a real vs. fake face classification task. Once I have a baseline that works on clean data, I will try to break it using two types of attacks. First, simple image degradations: JPEG compression at different quality levels, Gaussian noise, Gaussian blur, and downscale/upscale cycles. Second, proper adversarial attacks using FGSM and PGD. For each attack and intensity, I will track accuracy, precision, and recall, and plot how the detector degrades. After that, I will retrain with adversarial augmentation and see if the new model holds up better. If I have time, I also want to add Grad-CAM heatmaps to see where the detector is looking before and after attacks.

Implementation

For the dataset, I am looking at either a subset of FaceForensics++ or the 140k Real and Fake Faces dataset on Kaggle. It depends on what is easier to set up and work with. For the model, I will probably go with ResNet-18 or EfficientNet-B0 and fine tune the last layers. I have not locked in one yet and will decide after trying both out.

The attack pipeline will sweep over perturbation intensities for each attack type, like JPEG quality from 10 to 90, Gaussian noise with increasing sigma, and FGSM/PGD over a range of epsilon values. For each setting I will log the metrics so I can plot degradation curves. For the adversarial retraining, I will mix clean and adversarial examples into the training data and retrain from the pretrained checkpoint.

Tools: Python, PyTorch, OpenCV, `torchattacks`, `pytorch-grad-cam`. I have access to the university cluster for GPU training if needed.

Related Work

I have gone through a few papers in this area. Hou et al. at CVPR 2023 showed that attackers can exploit frequency and brightness statistics to get past detectors [1]. Gandhi and Jain at IJCNN 2020 showed that even basic FGSM/PGD attacks can drop detector accuracy significantly [2]. Neekhara et al. at CVPRW 2021 looked at how well adversarial attacks transfer across different detector architectures [3]. Goswami et al. proposed a loss function specifically designed to make deepfake detectors robust against adversarial inputs [4].

Expectations

I expect the baseline to do well on clean data but fall apart under perturbations. Adversarial training should help, but I am not sure by how much, especially against stronger attacks like PGD. I have not worked with adversarial ML before, so there is definitely a learning curve. If the defense does not end up helping much, that is still a valid result worth writing up.

References

[1] Y. Hou et al., "Evading DeepFake Detectors via Adversarial Statistical Consistency," *CVPR*, 2023.

[2] A. Gandhi and S. Jain, "Adversarial Perturbations Fool Deepfake Detectors," *IJCNN*, 2020.

[3] P. Neekhara et al., "Adversarial Threats to Deepfake Detection: A Practical Perspective," *CVPRW*, 2021.

[4] S. Goswami et al., "Adversarially Robust Deepfake Detection via Adversarial Feature Similarity Learning," *arXiv:2403.08806*, 2024.